

CS F364: Design & Analysis of Algorithms

Quiz 5 — June 12, 2026

Coverage: Lectures 1–14

Note

Instructions:

- Time: 15 minutes.
- Write your answers clearly on paper, scan, and upload to Google Classroom.
- Show all steps for full credit.

Question: LCS — Dynamic Programming

Let $c[i, j]$ be the length of the LCS of prefixes $X[1..i]$ and $Y[1..j]$.

For strings $X = \text{ABCD}$ and $Y = \text{AXCD}$:

- (a) [2 marks] Write the complete recurrence relation for $c[i, j]$ for all $i, j > 0$. Clearly state the base case and both cases ($x_i = y_j$ and $x_i \neq y_j$).
- (b) [2 marks] Fill in the DP table below. The first row ($i = 0$) and first column ($j = 0$) are already given.
- (c) [1 mark] What is the length of the LCS? What is the actual LCS sequence?

Solution

Part (a) — Recurrence Relation

Base case: For $i = 0$ or $j = 0$, $c[i, j] = 0$.

Recurrence: For $i, j > 0$:

$$c[i, j] = \begin{cases} c[i-1, j-1] + 1 & \text{if } x_i = y_j \\ \max(c[i-1, j], c[i, j-1]) & \text{if } x_i \neq y_j \end{cases}$$

$c[i, j]$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
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Part (b) — Completed DP Table

$c[i, j]$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
	$\emptyset$	A	X	C	D
$i = 0$ $\emptyset$	0	0	0	0	0
$i = 1$ A	0	1	1	1	1
$i = 2$ B	0	1	1	1	1
$i = 3$ C	0	1	1	2	2
$i = 4$ D	0	1	1	2	3

Computation walkthrough:

- **Row 1 (A):** $x_1 = A$, compare with y_j :
 - $j = 1$ (A): $A = A \Rightarrow c[1, 1] = c[0, 0] + 1 = 1$
 - $j = 2$ (X): $A \neq X \Rightarrow c[1, 2] = \max(c[0, 2], c[1, 1]) = \max(0, 1) = 1$
 - $j = 3$ (C): $A \neq C \Rightarrow c[1, 3] = \max(c[0, 3], c[1, 2]) = \max(0, 1) = 1$
 - $j = 4$ (D): $A \neq D \Rightarrow c[1, 4] = \max(c[0, 4], c[1, 3]) = \max(0, 1) = 1$
- **Row 2 (B):** $x_2 = B$, all y_j are A, X, C, D — none match B:
 - $c[2, j] = \max(c[1, j], c[2, j-1])$ — all cells = 1
- **Row 3 (C):** $x_3 = C$, matches $y_3 = C$:
 - $j = 1$ (A): $C \neq A \Rightarrow c[3, 1] = \max(c[2, 1], c[3, 0]) = \max(1, 0) = 1$
 - $j = 2$ (X): $C \neq X \Rightarrow c[3, 2] = \max(c[2, 2], c[3, 1]) = \max(1, 1) = 1$
 - $j = 3$ (C): $C = C \Rightarrow c[3, 3] = c[2, 2] + 1 = 1 + 1 = 2$
 - $j = 4$ (D): $C \neq D \Rightarrow c[3, 4] = \max(c[2, 4], c[3, 3]) = \max(1, 2) = 2$
- **Row 4 (D):** $x_4 = D$, matches $y_4 = D$:
 - $j = 1$ (A): $D \neq A \Rightarrow c[4, 1] = \max(c[3, 1], c[4, 0]) = \max(1, 0) = 1$
 - $j = 2$ (X): $D \neq X \Rightarrow c[4, 2] = \max(c[3, 2], c[4, 1]) = \max(1, 1) = 1$
 - $j = 3$ (C): $D \neq C \Rightarrow c[4, 3] = \max(c[3, 3], c[4, 2]) = \max(2, 1) = 2$
 - $j = 4$ (D): $D = D \Rightarrow c[4, 4] = c[3, 3] + 1 = 2 + 1 = 3$

Part (c) — LCS

LCS length: $c[4, 4] = 3$

LCS sequence: ACD

(Traceback: $c[4, 4]$ came from $c[3, 3] + 1$ where $D = D$; $c[3, 3]$ came from $c[2, 2] + 1$ where $C = C$; $c[2, 2]$ came from $c[1, 1]$ (max); $c[1, 1]$ came from $c[0, 0] + 1$ where $A = A$. Reading the matched characters in reverse: D, C, A → “ACD”.)